

Lab Assignment 1 and 2

Lab Assignment 1 - Explore OpenAI Gymnasium

1. Write Python code to reset the 3D Tic-Tac-Toe environment and display the initial state.
2. Write Python code to display the state space and action space of the 3D Tic-Tac-Toe environment.
3. Write Python code to select a random action, execute the action, and display the new state and reward.
4. Write Python code to execute 5 random actions and display the selected action and reward after each action.
5. Write Python code to check and display whether the episode is terminated or truncated after executing an action.

Lab Assignment 2 - Reinforcement Learning using Tic-Tac-Toe

1. Write Python code to train the Tic-Tac-Toe agent using 100 and 1,000 training episodes and compare the agent's performance.
2. Write Python code to train the agent using two different numbers of training episodes and compare the number of wins, losses, and draws.
3. Write Python code to run the trained agent against a random player and display the game result.
4. Write Python code to compare the agent's behavior before and after training.
5. Write Python code to modify one training parameter and observe its effect on the agent's learning performance.